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PI Name:	Pluth, Janice Ph.D.		
Project Title:	Space Adaptation Effects on Immune System Impacts Reproductive Function and Mammary Gland Development Across Generations		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Developmental biology		
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Space Biology Cross-Element Discipline:	(1) Reproductive Biology (2) Developmental Biology (3) Immunology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
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Zip Code:	89154-9900	**Congressional District:**	1
Comments:	NOTE: Formerly at Lawrence Berkeley National Laboratory until fall 2017.		
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No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA ARC
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COI Name (Institution):	Snijders, Antoine Ph.D. (Lawrence Berkeley National Laboratory) Sridharan, Deepa Ph.D. (Lawrence Berkeley National Laboratory)		
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Task Description:

Space travel results in adaptation to unique conditions that include exposure to low doses of cosmic radiation and microgravity, and heightened stress. Adapting to these physiological and psychological stressors can alter hormone and immune functions, disrupting normal development of hormone-sensitive organs and negatively impacting reproductive health. Studies investigating mammary gland development in young adult mice will help define how adaptations to space travel can influence these fundamental processes. The ductal network of the mammary gland contains a number of different cell types including epithelial cells, adipocytes, immune cells, vascular endothelial cells, and fibroblasts. Signaling between these cell types is highly sensitive to environmental perturbations. In addition, both hormone and immune responses have been shown to play a critical role in ductal network branching of the developing mammary gland. A combination of novel assays will be used to understand how space exposures impact immune function adversely affecting mammary gland development, stem cell populations, and the reproductive ability of mice. We will test whether space adaptation-induced epigenetic changes in the oocytes of mice correlate with reproductive ability. We hypothesize that the space environment weakens the immune system, and alters hormone balance resulting in alterations in number and function of mammary stem cells and development of the mammary gland. Additionally, we hypothesize that epigenetic changes as a result of space exposure will affect reproductive capability as compared to mice on Earth. We further propose that the sensitivity to these effects is dependent on the genetic background of exposed mice. These studies will aid in defining how alterations in immune function from adaptation to space travel and subsequent re-adaptation on Earth, result in defects in organ development, reproductive ability, and epigenetic changes affecting the fitness of subsequent generations.
July 2016: Definition stage, ground-based experiments to prepare for in-flight space studies. Hypothesis for ground-based study: We will test the hypothesis that the genetically driven sensitivity of the immune system affects mammary gland development post fractionated Low LET (low energy transfer) radiation exposures. In space flight, mice are subject to chronic low-dose radiation exposures. This low dose of radiation exposure may have a significant impact on endocrine function and immune response and influence organ development. Prior to in-flight experiments it would be useful to perform ground-based studies using fractionated X-ray exposures to streamline procedures and to obtain information regarding how genetic background and associated immune system response impacts mammary gland development following radiation exposures. To answer this, we propose to test our hypothesis with the following specific aims in these modified ground-based studies:

AIM 1: To determine how low-dose fractionated radiation affects immune function in young adult mice and its consequences on mammary organ development.

AIM 1.1: To examine acute effects of immune response to low-dose fractionated radiation on mammary gland development.

AIM 1.2: To investigate how genetic background modulates the biological effects of fractionated radiation.

EDITOR'S NOTE (December 2017; updated April 2018): Project is being continued by "Space Adaptation Effects on Immune System Impacts Reproductive Function and Mammary Gland Development Across Generations--80NSSC18K0373/80NSSC18K0019" with the same PI, Dr. Janice Pluth, due to Principal Investigator's move to University of Nevada, Las Vegas.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

This ground-based study was initiated to provide relevant data for subsequent inflight experiments to follow. The intent of the original inflight proposal was to define the effects of space exposures on organ development. However it was recently noted that inflight studies would not be possible, so we have modified the proposal to complete ground-based studies in such a manner as to have a published piece of work from these studies. In these studies we have compared the effect of fractionated radiation vs acute exposure on immune response, stem cell populations, and mammary gland development using two genetically diverse strains of mice. One strain of mice (BALB/c) has a greater sensitivity to radiation than the other strain (C57BL/6). The comparison of these two strains will allow us to define whether there is a genetic susceptibility to radiation effects. In space flight, mice will be subject to a chronic low dose exposure to radiation. This low dose of radiation may have a significant impact on endocrine function and immune response and influence organ development. Low dose fractionated exposures here on Earth could provide some insights as to how the protracted low doses received in space might affect organ development. Thus prior to in-flight experiments we will be able to obtain information regarding how genetic background and associated immune system response impact mammary gland development following fractionated or acute low LET radiation exposures.
Materials & Methods: Two genetically unique mice strains, BALB/c and C57BL/6 were used for these studies. Animals were obtained at 3 weeks of age and provided 1 week to equilibrate prior to studies. Mice were irradiated at 4 weeks of age, either fractionated (5 daily fractions of 10 cGy) or one acute dose (50cGy). Following the last fraction or acute exposure mice were left to continue growing until 7 weeks of age, at which point they were sacrificed. Blood was collected by orbital bleeds at the time of sacrifice into EDTA-containing eppendorf tubes. A portion of the blood was placed in Streck fixative for subsequent FACS analysis of T and B cell populations. Another portion was used to define CBC and differential, and lastly a portion was centrifuged and plasma collected and stored frozen (-80°C) to use to define cytokine profiles. Two cohorts of each strain were irradiated to obtain mammary glands from one for LSTAFI (Laser Scanning based Tissue Autofluorescence/ Fluorescence Imaging) and immune infiltration studies, and the second cohort's mammary glands were used for stem cell characterization. Assays below detail those that we have performed thus far. A flow cytometry-based assay was used to quantify B and T cell changes. A Complete Blood Cell Count (CBC) and differential were obtained from all mice sacrificed in the study. For these ground-based studies, the Hemavet HV950FS (Drew Scientific) was used to obtain a five-part complete white blood cell differential. Mammary gland whole mounts were made for one cohort of the mice and gland branching was quantified bid LSTAFI. Consequences of fractionated vs acute radiation on mammary architecture, including branching and vascularization /angiogenesis was assessed using this novel imaging approach, which allows for 3-D visualization of macro- and micro-anatomy of samples without the need for tissue sectioning or staining. These images are analyzed using Imaris 3D imaging analysis software to determine presence of defects in mammary development including ductal hyperplasia, and development of premature terminal end buds.

Brief summary of past year's research accomplishments. Our first cohort of animals were irradiated in July of 2016 (36 animals) and second cohort in June of 2017 (24 animals). At the 7 week time period following radiation or mock

	irradiation exposure at 4-5 weeks, animals were sacrificed and mammary glands dissected out for LSTAFI imaging of the whole gland, immune infiltration studies, and mammary stem cell profiles. In addition, blood was also obtained from all of the animals for CBC and differential, B and T cell profiles, and cytokine studies. Irradiated and mock-irradiated mice did not show gross differences in appearance following radiation exposures.
Task Progress:	Previously we noted that BALB/c mice showed a decreased weight gain upon both types of radiation regimen exposures, whereas C57BL/6 mice show increased weight gain. For both strains of mice ductal growth and weight gain failed to show a lack of correlation in irradiated mice whereas in control mice there was a nice correspondance. This latter radiation effect was evident with both fractionated and acute exposures. Radiation regimen and strain dependent differences in blood and immune cell populations were also evident, with BALB/c mice typically showing a more effect as compared to C57BL/6 mice. Studies detailing defects in the mammary gland using LSTAFI revealed mammary ductal branching as well as the size and number of terminal end buds (TEB) in the mammary gland differed based on the radiation regimen of exposure. Of particular note, an increase in TEB size and number upon fractionated radiation exposure was observed in both strains. Since the last task report we have performed additional analysis of 40X images of the TEBs from each stain and exposed population. From this analysis we observe changes that are common between strains that are radiation regimen dependent, but also unique changes specific to the mouse strain. For example both strains show an increase in the TEB diameter with acute radiation and an increase in diameter with fractionated radiation. In addition, both strains show a decrease in the width of the neck region with either fractionated or acute radiation exposure. More fine detailing analysis reveals fractionated radiation to increase the body cell top and cap layer of cells but to decrease the number of cells growing on the side of the body cell of the TEB in BALB/c mice as compared to controls. This may indicate that acute exposures are preventing the differentiation of stem cells to correctly differentiate and form the ducts. This latter pattern does not appear evident in the C57BL/6 mice. We have also stained mammary gland tissue to investigate immune cell infiltration into the gland plus and minus different radiation regimen exposures. We are currently in the process of quantifying these changes but have observed changes as detected by eye that appear specific to radiation regimen of exposure. We will quantify these changes once funds have been transferred to University of Nevada Las Vegas (UNLV) and we can again fund the continuation of this work. The unique patterns of immune infiltration in conjunction with the defects in the mammary gland will aid in providing a more complete picture regarding how radiation exposure during mammary gland development affects growth and development and the importance of genetics and the immune response. For future work we would like to complete the work proposed which includes radiation exposure followed by a challenge with LPS to define how immune system function may be compromised by the different radiation regimen exposures. Initiation of this final phase of work will be started once funds are successfully transferred from LBNL (Lawrence Berkeley National Laboratory) to UNLV. These studies will provide crucial information regarding how genetics in combination with radiation may impact the immune system and disturb the growth and development of an organ. Using very low doses of fractionated radiation will help to mimic the low dose chronic radiation in space and provide speculative information regarding effects that may be achieved in space. EDITOR'S NOTE (December 2017; updated April 2018): Project is being continued by "Space Adaptation Effects on Immune System Impacts Reproductive Function and Mammary Gland Development Across Generations--80NSSC18K0373/80NSSC18K0019" with the same PI, Dr. Janice Pluth, due to Principal Investigator's move to University of Nevada, Las Vegas. See that project for subsequent reporting.
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